

TACHENG, China

WMO: 511330

Lat: 46.7320N

Lon: 82.9798E

Elev: 535

StdP: 95.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-20.8	-27.5	0.3	-22.6	-25.0	0.4	-19.9	8.4	-5.5	7.1	-5.3	1.4	20	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	34.4	18.5	32.9	17.9	31.3	17.5	20.1	30.2	19.3	29.2	18.6	28.2	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.9	22.8	16.0	12.1	22.6	15.1	11.4	22.6	60.2	30.2	57.3	29.2	54.9	28.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.3	5.1	-26.5	37.6	3.2	1.6	-28.8	38.8	-30.7	39.7	-32.4	40.6	-34.7	41.8		
(5)			-26.8	21.4	3.1	1.4	-29.0	22.4	-30.8	23.2	-32.6	24.0	-34.9	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-9.9	-6.7	1.5	11.1	16.9	22.1	23.7	22.5	16.9	9.3	0.6	-7.0	(6)
(7)		DBStd	12.75	5.36	5.25	6.24	5.08	4.17	3.33	2.90	3.34	4.48	4.84	5.93	5.22	(7)
(8)		HDD10.0	2289	616	468	268	47	3	0	0	0	4	72	286	526	(8)
(9)		HDD18.3	4082	874	701	521	219	76	8	1	6	77	281	533	784	(9)
(10)		CDD10.0	1740	0	0	6	81	217	362	424	387	212	50	2	0	(10)
(11)		CDD18.3	491	0	0	0	3	32	119	166	134	34	1	0	0	(11)
(12)		CDH23.3	5352	0	0	0	47	380	1230	1778	1503	398	15	0	0	(12)
(13)		CDH26.7	2194	0	0	0	7	113	496	798	656	123	1	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	1.9	2.4	2.9	2.6	2.3	2.1	2.1	2.2	2.1	2.0	1.7	(14)
(15)	Precipitation	PrecAvg	281	16	13	18	31	29	27	31	16	13	32	33	21	(15)
(16)		PrecMax	468	43	33	64	88	80	121	92	56	49	75	94	52	(16)
(17)		PrecMin	154	1	2	3	0	4	9	2	0	0	1	8	3	(17)
(18)		PrecStd	75	10	8	14	24	20	21	25	14	10	20	23	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	6.9	19.9	27.7	32.6	34.9	37.0	36.4	32.8	25.4	17.1	5.1	(19)
(20)			MCWB	0.1	2.0	9.8	12.9	16.3	18.3	19.6	18.7	17.3	12.3	8.6	1.1	(20)
(21)		2%	DB	0.8	4.1	16.2	24.5	29.6	33.1	34.7	34.1	29.8	22.2	13.5	2.6	(21)
(22)			MCWB	-1.2	0.3	8.0	11.9	14.9	17.9	19.0	18.3	15.4	11.0	6.8	-0.3	(22)
(23)		5%	DB	-0.8	2.2	13.3	22.2	27.4	31.5	32.9	32.3	27.8	19.7	10.4	1.1	(23)
(24)			MCWB	-2.8	-0.8	6.8	10.8	14.3	17.4	18.3	17.7	14.5	10.0	5.2	-1.0	(24)
(25)		10%	DB	-2.6	0.5	10.4	19.9	25.2	29.8	31.1	30.5	25.4	17.1	7.9	-0.4	(25)
(26)			MCWB	-4.4	-1.9	5.4	10.0	13.5	17.0	18.0	17.1	13.5	9.1	4.2	-2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.5	10.6	14.6	17.6	20.4	21.9	20.7	17.9	13.4	9.8	1.9	(27)
(28)			MCDWB	2.8	5.7	17.7	23.9	28.8	30.4	33.4	31.7	29.8	21.6	14.7	3.9	(28)
(29)		2%	WB	-0.7	0.8	8.8	13.1	16.4	19.4	20.5	19.6	16.4	11.9	7.5	0.4	(29)
(30)			MCDWB	0.5	3.3	14.9	21.0	25.6	29.6	30.6	30.2	27.8	20.0	11.7	1.9	(30)
(31)		5%	WB	-2.7	-0.5	7.3	12.0	15.5	18.6	19.6	18.6	15.3	10.8	5.9	-0.8	(31)
(32)			MCDWB	-1.1	1.9	12.4	19.4	23.9	28.3	29.1	29.2	25.6	18.0	9.7	0.8	(32)
(33)		10%	WB	-4.3	-1.8	5.6	11.0	14.6	17.8	18.9	17.8	14.2	9.7	4.4	-2.1	(33)
(34)			MCDWB	-2.6	0.4	10.3	17.8	22.6	27.0	28.1	28.1	23.7	16.0	7.8	-0.4	(34)
(35)	Mean Daily Temperature Range	MDBR	9.1	9.5	9.4	11.2	11.8	12.1	12.4	13.2	13.0	11.2	8.4	8.1	(35)	
(36)		5% DB	MCDBR	9.7	10.8	12.8	14.6	14.7	14.4	15.0	15.6	15.5	15.0	12.1	8.8	(36)
(37)			MCWBR	7.6	7.9	6.9	6.4	5.6	5.2	5.3	6.0	6.7	7.1	6.8	6.5	(37)
(38)		5% WB	MCDWB	8.6	9.6	11.8	11.7	12.1	12.5	12.7	13.4	14.2	13.1	10.9	7.6	(38)
(39)			MCWBR	6.9	7.2	6.6	5.7	5.2	4.9	5.0	5.5	6.6	6.6	6.6	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.338	0.385	0.425	0.406	0.411	0.414	0.385	0.358	0.356	0.339	0.307	(40)	
(41)		taud	2.089	2.060	2.058	2.065	2.183	2.236	2.250	2.308	2.335	2.239	2.160	2.100	(41)	
(42)		Ebn at Noon	747	803	823	827	859	855	846	858	851	781	704	694	(42)	
(43)		Edh at Noon	94	120	140	154	142	135	132	119	106	99	88	83	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.86	2.92	4.10	5.23	6.51	6.87	6.59	6.04	4.75	2.94	1.66	1.38	(44)	
(45)		RadStd	0.11	0.11	0.26	0.34	0.33	0.35	0.41	0.34	0.28	0.24	0.15	0.11	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page